

Implementation of a pharmacist-led clinic for hypertensive patients in primary care – a pilot study

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Key words

Antihypertensive agent
Hypertension clinic
Ischaemic heart disease
Pharmaceutical care
Practice guideline
Primary medical care
Primary prevention
Scotland
Secondary prevention

Abstract

Objectives: To implement a pharmacist-led Hypertension Management Clinic in one general medical practice. To evaluate the impact of the clinic on blood pressure (BP) control and prevention of coronary heart disease (CHD).

Method: A total of 242 patients attended the pharmacist-led hypertension clinic over a 10-month period. Lifestyle and drug therapy alterations were implemented to achieve British Hypertension Society (BHS) target level BP. A sub-group of 160 patients were used to compare BP control in the clinic setting against that with the general practitioner (GP). Assessment was made of 10-year CHD risk in patients with no atherosclerotic disease. Patients with underlying atherosclerotic disease were prescribed statins, and antiplatelet drugs where indicated.

Main outcome measures: Changes in numbers of hypertensive patients meeting the BHS target level BP. Changes in prescribing of antiplatelet agents and statins for primary and secondary prevention of atherosclerosis.

Results: In 206 patients with established hypertension, the number achieving target level BPs increased from 74 (36%) pre-clinic to 174 (85%) post-clinic; $P < 0.001$ chi-squared test. After attending the clinic, for 5 months 74 patients (80%) achieved target level BP in the clinic compared with 27 (40%) with standard GP care; $P < 0.001$ chi-squared test. Of 188 patients assessed for primary prevention therapy, 126 (67%) required treatment with aspirin and 37 (20%) with a statin. Post-clinic 101 (80%) received aspirin compared with 17 (13%) pre-clinic and 34 (92%) received a statin in comparison with 4 (11%) pre-clinic; both $P < 0.001$ chi-squared test. A total of 52 (96%) of 54 patients received an antiplatelet agent for secondary prevention of atherosclerosis compared with 40 patients (74%) pre-clinic. Thirty six of 54 patients required a statin for secondary prevention. Thirty five patients (97%) received a statin compared with 23 (64%) pre-clinic; both $P < 0.01$ chi-squared test.

Conclusion: Implementation of a pharmacist-led clinic improved blood pressure control and appropriate prescribing of antiplatelet agents and statins for primary prevention of CHD and secondary prevention of atherosclerosis.

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Introduction

Guidelines were published for the management of hypertensive patients in the UK which recommended strategies be introduced in primary care for implementation^{1,2}. The problem of how to resource implementation of these guidelines within primary care has been the subject of debate within the medical literature^{3–5}. The guidelines define target and audit standard levels for blood pressure (BP) in the clinic setting, which

are shown in Table 1¹. The audit standard level is the minimum recommended level of BP control for individual patients and general practices will be remunerated for achieving this BP under the new UK General Medical Services (GMS) contract⁶.

The evidence base for reducing morbidity and mortality secondary to hypertension with drug therapy is well established^{7,8}. Lifestyle alterations have also been shown to significantly improve both systolic and diastolic BP and it is important these are addressed^{9–13}.

Population surveys have shown that 50 to 75% of patients diagnosed with hypertension in North America, Europe and Australia do not achieve target levels¹⁴. In one hypertension treatment trial, of the 20,000 patients enrolled, 50% had uncontrolled BP levels despite treatment¹⁵. Most patients require more than one medicine in conjunction with lifestyle changes to achieve target BP levels¹⁶.

Although it is important that hypertension is managed effectively, it is an indicator of cardiovascular risk, and treatment of all other risk factors should be effected. A number of risk assessment tools have been devised, based on the Framingham model¹⁷, to assess cardiovascular risk including the new Sheffield table¹⁸, the New Zealand guidelines¹⁹ and the Joint British recommendations².

Involvement of pharmacists in the management of patients with hypertension has been shown to improve BP control in several small studies where older target ranges were used^{20–23}. No published studies have looked at pharmacists' involvement in overall risk management of this patient group. The National Health strategy for Scotland promotes chronic disease management by pharmacists²⁴.

Method

Patient identification and data collection

All adult patients with a diagnosis of essential hypertension, who were coded on the practice computer, were considered for inclusion. Since morbidity coding data in the practice was incomplete at clinic inception, patients were also identified for inclusion through a search using bendrofluazide. Bendrofluazide was chosen since it was the most commonly prescribed antihypertensive agent within the practice and is rarely prescribed for other clinical indications. All medical notes were screened by the pharmacist to confirm the diagnosis. Patients were randomised sequentially, prior to study inception, by the pharmacist into two groups. The exclusion criteria are detailed in Table 2.

Patients who attended the Hypertension Management Clinic for the duration of the study, from August 2001 until May 2002, were designated as Group 1, the active group. The control group, Group 2, received standard treatment through their general practitioner (GP) for the initial 5 months and were invited to attend the Hypertension Management Clinic from January

Table 1 Recommended target level and audit standard BP

Blood pressure	No diabetes	Diabetes
Target level	<140/85 mmHg	<140/80 mmHg
Audit standard	<150/90 mmHg	<140/85 mmHg

Adapted from Ref. 1.

Table 2 Hypertension Management Clinic – patient exclusion criteria

Patients incorrectly coded as hypertensive
Patients with dementia or who were housebound
Patients with hypertension already being managed by secondary care specialists (e.g. diabetic patients attending diabetic clinics or patients attending hypertension clinics)
Patients with hypertension secondary to pregnancy
Patients considered unsuitable for inclusion by their GP

2002. This is illustrated in Figure 1. Hypertensive patients not coded as such could also be referred directly to the clinic by their GP.

Patients were categorised depending on the length of time they had been treated for hypertension to allow appropriate assessment of 10-year coronary heart disease (CHD) risk^{18,25–27}. Existing patients were those treated with anti-hypertensive medicines for greater than one year, all other patients were designated newly diagnosed patients.

Relevant patient characteristics were collected including duration of hypertension, relevant co-morbidities, total number of anti-hypertensive medicines taken and BP on admission to the study. The characteristics of Groups 1 and 2, identified from the computer search were compared.

Implementation of Hypertension Management Clinic

A treatment protocol was devised based on the guidelines^{1,2}. The Joint British recommendations

were used to calculate 10-year CHD risk for newly diagnosed patients since evidence in UK populations' supports that this is the most accurate^{25,26}. The new Sheffield table was used to estimate cardiovascular risk in patients treated with anti-hypertensive medication because of its applicability in this patient group^{18,27}. The pharmacist discussed all changes to prescribed medication with the patient and their GP, prior to alteration. Dose titration was undertaken by the pharmacist without GP consultation. Details of the consultation including lifestyle modification advice were documented in the patient records. Changes in medication were entered on the practice computer system and prescriptions were signed by a GP. Blood samples required to monitor treatment or evaluate cardiovascular risk were taken by the pharmacist or nursing staff and patients requiring an electrocardiogram were referred to nursing staff. All patients had their urine tested for glucose and protein. The clinic ran for 2-h sessions, 4 days per week and patients were allocated 15-min appointments. Patients attended the clinic at intervals of 2 weeks to 3 months depending on BP control¹.

Evaluation of clinical impact

Blood pressure management

The total number of patients meeting the audit standard and target level for BP on entry were quantified and compared with those meeting the target and audit standard following management within the clinic.

Primary and secondary prevention of CHD

Patients with no underlying arteriosclerotic diseases were assessed, following completion of the appropriate tests, for their 10-year CHD risk on entry to the study. Patients requiring treatment with statin or antiplatelet therapy as primary prevention were identified. A comparison was made of the proportion of patients appropriately receiving primary prevention therapy before and after clinic attendance. Number of patients receiving treatment with anti-platelet agents or statins as secondary prevention therapy due to arteriosclerosis was contrasted before and after inclusion in the study.

Effect of redesigned service

The effects of the redesigned service was measured by comparing the number of patients meeting target range and audit standard BP in group 1, following 5 months in the clinic setting, with those in group 2, who received care via their GP.

Pharmaceutical care issues addressed

Pharmaceutical care issues identified and addressed within the clinic were collated and categorised according to Hepler and Strand's criteria²⁸.

Assessment of patient opinion

A semi-structured questionnaire was designed to assess patients' opinions of the clinic (Appendix 1). The questionnaire was posted once to the first 150 patients who attended the clinic and a prepaid envelope was enclosed for return to the Trust Clinical Gover-

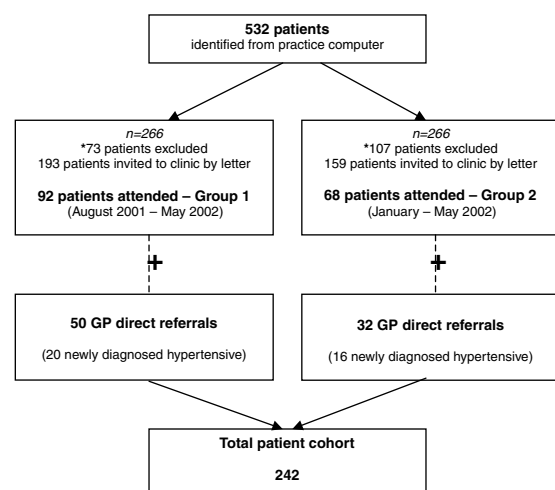


Figure 1 Flow diagram of patients attending the Hypertension Management Clinic. *Table 2 – Patients exclusion criteria.

Table 3 Comparison of baseline patient characteristics

	Total patients (n = 242)	Group 1 (n = 92)	Group 2 (n = 68)
Mean age (SD) years	65 (10)	66 (11)	67 (10)
Male	111 (45.9%)	44 (47.8%)	26 (38.2%)
Female	131 (54.1%)	48 (52.2%)	42 (61.8%)
Mean blood pressure (SD) mmHg at entry	Existing patients (n = 206) Systolic 148 (19.1) Diastolic 79 (11.0)	Systolic 147 (17.6) Diastolic 80 (10.2)	Systolic 145 (17.6) Diastolic 78 (10.5)
Median (IQR) duration of hypertension years	Existing patients (n = 206) 6 (3,10.25)	7 (3,12)	6 (3,9)
Mean BMI (SD) kg/m ²	29.3 (5.3)	29.2 (5.4)	29.6 (5.0)
Primary prevention patients	*188 (77.7%)	70 (76.1%)	55 (80.9%)
CHD risk ≥ 30% CHD	47 (19%)	17 (18.5%)	6 (8.8%)
risk ≥ 15% and <30%	79 (33.1%)	31 (33.7%)	25 (36.8%)
CHD risk <15%	59 (24.4%)	22 (23.9%)	24 (35.3%)
Secondary prevention patients	54 (22.3%)	22 (23.9%)	13 (19.1%)
Diabetes	22 (9.1%)	9 (9.8%)	6 (8.8%)
Positive family history of coronary heart disease	73 (30.2%)	25 (27.8%)	27 (39.7%)
**Left ventricular hypertrophy	34 (14%)	14 (15.2%)	7 (10.3%)
Current smokers	46 (19%)	15 (16.3%)	10 (14.7%)

SD, standard deviation; IQR, interquartile range.

* In three patients CHD risk could not be assessed due to patient not attending to relevant tests.

** In 126 patients LVH status was unknown since they had no documented ECG in their medical notes or had not had an ECG.

nance Department for collation. The questionnaires were not coded to preserve patient anonymity.

Data analysis

A Microsoft Access database was designed to facilitate data collection. Patient characteristics data were tested for Gaussian distribution and if asymmetrically distributed, is expressed as ordinal data. Baseline characteristics of groups 1 and 2 were compared using a chi-square test for nominal data, an unpaired Mann–Whitney *U*-test for ordinal data and an unpaired Student's *t*-test for interval data, all at 95% probability.

Comparison of patients attaining the target level and audit standard BP, and prescription of appropriate primary and secondary prevention therapy, pre- and post- clinic attendance was made using a chi-squared test, with 95% probability. The number of anti-hypertensive medicines prescribed per patient was compared using a Mann–Whitney *U*-test.

Results

Patient identification and characteristics

A total of 532 patients were considered for inclusion in the study from the practice computer system.

Group 1 (n = 92)

Of 266 patients identified, 73 were excluded. A total of 193 patients were invited to attend the clinic of whom 92 (47.7%) attended.

Group 2 (n = 68)

Of 266 patients identified, 107 were excluded. A total of 159 patients were invited to attend the clinic of whom 68 (42.8%) attended.

Fifty patients were referred directly to the clinic by their GP between August and December 2001 of whom 20 were newly diagnosed hypertensives. A further 32 patients were referred between January and May 2002 of whom 16 were newly diagnosed. Thus the total number of patients attending the clinic was 242 (Figure 1).

Patient characteristics

Baseline characteristics for the 242 patients attending the clinic are detailed in Table 3. There were no statistically significant differences between groups 1 and 2.

Evaluation of clinical impact

Blood pressure management

Existing patients (n = 206). Significant changes in the number of existing patients meeting both the audit standard and target level BP were made. Pre-clinic 115 patients (55.8%) met the audit standard BP compared with 196 (95.1%) after clinic attendance, $p < 0.001$. Pre-study 74 patients (35.9%) achieved their target level BP compared with 174 patients (84.5%) at the end of the study period, $P < 0.001$.

The number of anti-hypertensive medicines taken by patients did not differ significantly in the existing patients before and after attending the clinic. Median (IQR) was 2 (1,2) pre-study and 2 (1,2) post-study, $P > 0.2$.

Newly diagnosed patients ($n = 36$). Two patients did not require treatment for hypertension with medication and received lifestyle advice only¹. Of the remaining 34 patients, 12 (35.3%) achieved the audit standard BP pre-study compared with 32 (94.1%) on study conclusion, $P < 0.001$. Seven patients (20.6%) attained their target level BP prior to the study in comparison with 30 patients (88.2%), $P < 0.001$.

Primary and secondary prevention of CHD

Primary prevention of CHD. Significant changes in prescribing of aspirin and statins were made. The results are shown in Table 4.

According to the guidelines, 126 patients (67%) assessed for primary prevention therapy were eligible for treatment with aspirin. Twenty patients received aspirin pre-study including patients purchasing aspirin. Three of these patients were inappropriately prescribed aspirin therapy and it was discontinued. Aspirin was prescribed in an additional 84, thus the total number of patients receiving aspirin at the end of the study was 101 (80%) $P < 0.001$. Twenty-five patients were not taking aspirin at the end of the study despite having an indication. Of these, 15 patients had demonstrated gastrointestinal intolerance or allergy to the drug and five patients refused treatment.

Forty-seven patients had 10-year CHD risk of $\geq 30\%$ and were assessed for statin therapy. Of these, 31 patients had a cholesterol level of ≥ 5 mmol/l and were under the age of 70 years and six patients had familial hyperlipidaemia. On entry to the study, eight patients received a statin, which was appropriately prescribed in four patients. On study conclusion, 34 patients (92%) were appropriately receiving a statin, $P < 0.001$.

Secondary prevention of CHD. Significant changes were also seen in the prescribing of secondary prevention therapy after clinic attendance (Table 4). All 54 patients with established atherosclerotic disease

should have been receiving anti-platelet therapy. The term atherosclerotic disease includes patients with ischaemic heart disease, peripheral vascular disease and cerebrovascular disease. On entry to the clinic 40 patients (74%) received aspirin, which increased to 51 with one patient being prescribed clopidogrel (96%), $P < 0.01$. Of the 54 patients, 36 required treatment with a statin. Prior to clinic inception, 23 (64%) appropriately received a statin according to guidelines, which rose to 35 patients (97%), $P < 0.01$.

Effect of redesigned service

Seventy-four patients (80.4%), in group 1, had attained their target level blood pressure following 5 months attendance at the clinic and 88 patients (95.7%) had achieved the audit standard BP. In comparison, 27 patients (39.7%) from group 2 were at target level and 40 patients (58.8%) had achieved the audit standard BP through standard care via their GP. There is an improvement in the number of patients meeting both the audit standard and target level BP, $P < 0.001$. This is illustrated in Table 5.

Pharmaceutical care issues addressed in the study population

Over 500 pharmaceutical care issues were addressed during the study period mainly relating to medication additions and alterations for control of BP and addition of anti-platelet agents and statins for primary and secondary prevention of CHD. The service delivered was holistic and patients medication was also adjusted for control of angina, diabetes, heart failure, pain, hypothyroidism and gout. Over 600 biochemistry screens were ordered to assess 10-year CHD risk, monitor drug therapy appropriately and identify adverse effects. Eight newly diagnosed non-insulin dependent diabetic patients were identified through the clinic. These patients were given dietary advice and referred to their GP.

Assessment of patient opinion

Of the 147 patients sent the questionnaire, 110 completed questionnaires (74.8%) were received by the 3-week deadline. One hundred and seven patients (97%) specified that they wanted the pharmacist-led clinic to continue and 84 (76%) responders felt their understanding of hypertension increased through attending the clinic. The advice from the pharmacist on medicines was rated as good or very good by 103 (94%), 75 patients (68%) acknowledged receipt of advice on lifestyle changes from the pharmacist and of

Table 4 Changes in prescription of primary and secondary prevention medicines

	Pre-clinic Post-clinic		
Primary prevention	$n = 188$		
No. of patients receiving aspirin (%) $n = 126$	17 (13.5)	101 (80.2)	$P < 0.001$
No. of patients prescribed a statin (%) $n = 37$	4 (10.8)	34 (91.9)	$P < 0.001$
Secondary prevention	$n = 54$		
No. of patients receiving antiplatelet medicines (%) $n = 54$	40 (74.1)	52 (96.3)	$P < 0.01$
No. of patient prescribed a statin (%) $n = 36$	23 (63.8)	35 (97.2)	$P < 0.01$

Table 5 Effect of redesigned service on blood pressure control

	Group 1 ($n = 92$)	Group 2 ($n = 68$)	
No. of patients meeting target level (%)	74 (80.4%)	27 (39.7%)	$P < 0.001$
No. of patients meeting audit standard (%)	88 (95.7%)	40 (58.8%)	$P < 0.001$

those 59 (79%) said they made changes to their lifestyle. Eighty-nine patients (81%) thought it would be useful if the pharmacist could prescribe their antihypertensive medication.

Discussion

Blood pressure control within this patient population, prior to implementation of the clinic, is in line with published evidence that only half of all treated patients meet BP targets^{14,15}. In the study population, 56% of the existing patients were at the audit standard level on study initiation and 36% met the target level. Implementation of the pharmacist-led clinic demonstrated a significant and sustained improvement in blood pressure control over the study period within the patient cohort. Comparison of the redesigned service with the existing service for BP control suggests that the pharmacist-led clinic may be more effective at achieving target levels and audit standards in individual patients. This project was funded for a 1-year period primarily to assess if a pharmacist could contribute to improved patient care via development and delivery of a specialised clinic. It was not a rigorously conducted scientific study and several problems with the methodology are acknowledged. The patients should have been randomised into the 2 groups (groups 1 and 2) after study exclusion criteria had been applied to ensure equitable numbers in each group. In addition the results would be more valid if the active group had been compared to the control group over a 1-year period. Instead resources dictated that the comparison was made over a 5-month period and it is accepted that patients in the control group may not have attended their GP for a BP check and medication adjustment within this timeframe.

The estimated number of hypertensive patients in this practice is around 1000 patients, thus 25% of the total population attended the clinic. The clinic uptake by those patients invited to attend was disappointing at 45.5%. This may be explained by reluctance of patients to attend the clinic due to its novelty, since pharmacists did not routinely run clinics of this type in primary care in the UK. Attendance may also have been affected by the fact that the pharmacist-led clinic ran only in the afternoon because of problems with consultation space within the practice. It is a weakness of this study that the results cannot be applied to the general hypertensive population as a whole since BP control of patients who did not attend the clinic is unknown.

Within this patient cohort, significant improvements in prescribing of both primary and secondary prevention medicines for CHD have been demonstrated. Calculation of 10-year CHD risk is time consuming and evidence shows inaccuracies in the calculations made by health care professionals. A study in England reported that very few patients' medical records in primary care have the necessary data to make a risk assessment and that GPs are struggling with this workload which is reflected in the findings here²⁷.

The new GMS contract in the UK allows general practice income to increase through achievement of a

series of quality indicators supported by evidence-based clinical practice⁶. Hypertension management, stroke, diabetes and CHD constitute four of a total of 10 diseases where GPs will be remunerated for achievement of the indicators. The extension of pharmacists' roles within primary care is one way to help achieve improved standards for patients as well as appropriate reallocation of GP workload. The redesign of service provided here endorses this, however, this study did not assess the effect on GP workload. In addition to producing improvement in BP control the clinic was well received by patients who attended. The questionnaire was returned by 75% of the patients' and of those who responded to the questionnaire, 97% were keen that the clinic continued. The clinic has continued to run within the practice and the service has been extended to another general practice.

The implementation of this specialised clinic for management of hypertensive patients has obvious resource implications both in terms of personnel and drug costs. One problem is the lack of trained personnel, pharmaceutical or nursing, available to undertake this type of work. It is anticipated that changes in remuneration for community pharmacists in the UK will provide opportunities for this issue to be addressed. The legal framework within the UK for pharmacist supplementary prescribing offers a means of extending the opportunity for individualisation of a patient's medication and this type of clinic has provided an ideal setting for implementation²⁹. This pilot study demonstrates how a pharmacist working in partnership with GPs can deliver quality chronic disease management.

Conclusion

This study has shown that a pharmacist-led hypertension clinic, working in partnership with the primary care team, can improve control of blood pressure and evidence based prescribing of medicines to reduce prevalence of cardiovascular disease.

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Appendix 1. Patient questionnaire

**Newbyres Medical Group
Hypertension Management Clinic
Patient Questionnaire**

Please tick the appropriate box for each question

1. What age are you?

☐ Under 40 years
☐ 41-60 years
☐ 61-80 years
☐ 80+ years
2. Approximately how long have you had high blood pressure?

☐ Under 2 years
☐ 2-5 years
☐ 6-10 years
☐ 10+ years
3. If you had a choice, what time would you prefer to attend the clinic? Please tick one only

☐ 4-6 pm (current time)
☐ Morning
☐ Early afternoon
☐ Other - please say when this is
4. When you attended the clinic, did the pharmacist change the dose or type (brand name) of your high blood pressure medication?

☐ High blood pressure medication dose was changed
☐ High blood pressure medication type was changed
☐ My medication was not changed
☐ Don't know
- 5a. When you attended the clinic, did the pharmacist prescribe medication to lower cholesterol or change your existing cholesterol medication dose or prescribe aspirin?

☐ Cholesterol lowering medication was prescribed for the first time
☐ My current cholesterol medication dose was changed
☐ Aspirin was prescribed for the first time
☐ None of my medication was changed
☐ Don't know
- 5b. If your dose or type of medication has been changed, did you get an adequate explanation for the changes from the pharmacist?

☐ Yes
☐ No
☐ Not applicable
6. Do you think that attending the hypertension clinic increased your understanding of high blood pressure and its treatment?

Yes ☐
No ☐

If Yes, please tell us what was most helpful in increasing your understanding.

.....

PTO > > >

7. How would you rate the advice about your high blood pressure medicines from the clinic pharmacist?

☐ Very good
☐ Good
☐ Average
☐ Poor
- 8a. Did you receive any advice from the clinic pharmacist about lifestyle changes such as weight loss, dietary changes, or smoking? Yes ☐ No ☐
- 8b. Did you find the lifestyle advice helpful? Yes ☐ No ☐
- 8c. Did you make any changes to your lifestyle following advice at the hypertension clinic? Yes ☐ No ☐

If Yes, please tell us what changes you made or tried.

.....

.....

Currently the clinic pharmacist cannot prescribe medicines for treatment of high blood pressure or high cholesterol because the prescription must be signed by a doctor. This sometimes causes a delay in patients receiving new medication.

- 9a. If there was a delay for you to receive the prescription for a new type of medication, was this a problem for you?

☐ Yes
☐ No
☐ My medication has not been changed
- 9b. Do you think it would be helpful to patients if the clinic pharmacist was able to prescribe your high blood pressure medication? Yes ☐ No ☐
- 10. Do you have any comments about the clinic being run by a pharmacist rather than a GP?

.....
- 11. Do you think the hypertension clinic at the practice should continue? Yes ☐ No ☐
- 12. Do you have any other comments or suggestions about the hypertension clinic?

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Thank you for completing the questionnaire.

Please return it in the enclosed FREEPOST envelope by Friday 10th May 2002.

Survey - 73

Clinical Governance Support Team

Page - 1

Survey - 73

Clinical Governance Support Team

Page - 2